

Point

From Number Theory to Algebraic Geometry and Logic

Session S2 — Modules, Tensor Products, and Exact Sequences

Complete lecture notes with proofs, examples, and exercises

Abstract

This session builds the *linear* layer of the course: modules, tensor products, exact sequences, Nakayama’s lemma, and flatness. These are the tools in which every later structure is expressed: sheaves are module-valued functors (S26–S28), fibers of a family are tensors $M \otimes_R k(\mathfrak{p})$ (S30), Tor and Ext measure the failure of exactness (S42–S45), and flatness is the algebraic meaning of “continuously varying fibers” behind étale maps (S30). The unifying thread: a module is a *family of vector spaces parametrized by the points of $\mathrm{Spec} R$* , and tensoring is *evaluating the family at a point or changing the base*.

Session plan (150 minutes)

Time	Block
0:00–0:10	Orientation: why modules — sheaves, fibers, and the linear layer
0:10–0:40	§1 Modules, homomorphisms, free modules, exact sequences
0:40–1:10	§2 Tensor products: construction, universal property, right-exactness
1:10–1:20	<i>Break</i>
1:20–1:45	§3 Nakayama’s lemma and its corollaries
1:45–2:10	§4 Flatness: definition, localization, preview of S30 and S42–S45
2:10–2:30	Exercises / wrap-up: tensor = evaluation at a point

Conventions

All rings are commutative with $1 \neq 0$; all modules are unital ($1 \cdot x = x$). A *local ring* is a pair $(R, \mathfrak{m})$ with a unique maximal ideal $\mathfrak{m}$. By R -module we mean what is elsewhere called a left/right module (commutativity makes them agree).

Contents

1	Modules and Exact Sequences	2
2	Tensor Products	3
3	Nakayama’s Lemma	3
4	Flatness	4
5	Exercises	5
	Where This Session Feeds the Course	5

1 Modules and Exact Sequences

Definition 1.1 (Module). An R -**module** is an abelian group $(M, +)$ with an action $R \times M \rightarrow M$, $(r, x) \mapsto rx$, satisfying $r(x + y) = rx + ry$, $(r + s)x = rx + sx$, $(rs)x = r(sx)$, and $1x = x$.

Example 1.2 (Modules everywhere). • Ideals and quotient rings of R are R -modules; so is R itself and R^n .

- Over a field k , a module is exactly a k -vector space.
- Over $\mathbb{Z}$, a module is exactly an abelian group.
- Over $k[x]$, a module is a vector space V equipped with a linear operator T (the action of x).
- Localizations $S^{-1}M$ are $S^{-1}R$ -modules (and R -modules).

Definition 1.3 (Homomorphism; submodule; quotient). An R -**linear map** (module homomorphism) $f : M \rightarrow N$ satisfies $f(x + y) = f(x) + f(y)$ and $f(rx) = rf(x)$. Its kernel and image are submodules; quotients M/N are modules; the set of R -linear maps is $\text{Hom}_R(M, N)$, itself an R -module.

Definition 1.4 (Free; finitely generated). M is **free** with basis $\{e_i\}$ if every $x \in M$ is uniquely $x = \sum r_i e_i$; then $M \cong R^{(I)}$. M is **finitely generated** (f.g.) if $M = Rx_1 + \cdots + Rx_n$ for some x_i .

Proposition 1.5 (Rank is well defined). If R is commutative and $R^m \cong R^n$ then $m = n$. Hence free modules have a well-defined rank.

Proof. Pick a maximal ideal $\mathfrak{m}$ (S1, Krull). Tensoring $R^m \cong R^n$ with the field $R/\mathfrak{m}$ gives $(R/\mathfrak{m})^m \cong (R/\mathfrak{m})^n$ as vector spaces (tensor commutes with direct sums, §2); dimensions force $m = n$. $\square$

Definition 1.6 (Exact sequence). A sequence $\cdots \rightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \rightarrow \cdots$ is **exact** at M_i if $\text{im } f_i = \ker f_{i+1}$; **exact** if exact everywhere. A **short exact sequence** (s.e.s.) is $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$, i.e. f injective, g surjective, and $\text{im } f = \ker g$ (so $C \cong B/f(A)$).

Example 1.7. • $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow \mathbb{Z}/2 \rightarrow 0$ is exact.

- $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$ for any ideal I .
- $0 \rightarrow A \rightarrow A \oplus C \rightarrow C \rightarrow 0$ (inclusion/projection) — the model of a *split* s.e.s.
- $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{\times 3} \mathbb{Z} \rightarrow 0$ is *not* exact at the middle.

Definition 1.8 (Split s.e.s.). A s.e.s. $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ **splits** if there is a **section** $s : C \rightarrow B$ with $g \circ s = \text{id}_C$ (equivalently a **retraction** $r : B \rightarrow A$ with $r \circ f = \text{id}_A$).

Proposition 1.9 (Splitting lemma). For a s.e.s. as above, TFAE: (a) a section s exists; (b) a retraction r exists; (c) $B \cong A \oplus C$ compatibly with f, g .

Proof. (a) $\Rightarrow$ (c): define $\Phi : A \oplus C \rightarrow B$, $(a, c) \mapsto f(a) + s(c)$, and $\Psi : B \rightarrow A \oplus C$, $b \mapsto (f^{-1}(b - s(g(b))), g(b))$; here $b - s(g(b)) \in \ker g = \text{im } f$ so f^{-1} makes sense (f injective). One checks $\Phi\Psi = \text{id}$ and $\Psi\Phi = \text{id}$. (c) $\Rightarrow$ (a),(b) is immediate from the direct-sum projections/inclusions. (b) $\Rightarrow$ (a): set $s(c) = b - r(b)$ for any b with $g(b) = c$; well-defined and a section. $\square$

2 Tensor Products

Definition 2.1 (Bilinear map). A map $\beta : M \times N \rightarrow P$ is *R-bilinear* if $\beta(m, -)$ and $\beta(-, n)$ are *R*-linear for each m, n .

Definition 2.2 (Tensor product: construction). Let F be the free *R*-module on the set $M \times N$, and K the submodule generated by $(m + m', n) - (m, n) - (m', n)$, $(m, n + n') - (m, n) - (m, n')$, and $(rm, n) - (m, rn)$. Define $M \otimes_R N := F/K$, with $m \otimes n$ the class of (m, n) . Elements are finite sums $\sum m_i \otimes n_i$.

Theorem 2.3 (Universal property). *For any R-module P, the map $\Phi \mapsto ((m, n) \mapsto \Phi(m \otimes n))$ is a bijection $\text{Hom}_R(M \otimes_R N, P) \cong \{R\text{-bilinear } M \times N \rightarrow P\}$.*

Proof. Given bilinear β , the freeness of F gives $\tilde{\beta} : F \rightarrow P$ with $\tilde{\beta}(m, n) = \beta(m, n)$; bilinearity says exactly $\ker \tilde{\beta} \supseteq K$, so $\tilde{\beta}$ descends to $\Phi : M \otimes N \rightarrow P$ with $\Phi(m \otimes n) = \beta(m, n)$. Uniqueness: simple tensors generate $M \otimes N$. Conversely $\Phi(m \otimes n)$ is bilinear in (m, n) because Φ is linear and $\otimes$ is bilinear by construction. The two passages invert each other. $\square$

Example 2.4 (First computations). • $\mathbb{Z}/m \otimes_{\mathbb{Z}} \mathbb{Z}/n \cong \mathbb{Z}/(m, n)$: tensor $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow \mathbb{Z}/m \rightarrow 0$ with $\mathbb{Z}/n$; right-exactness (Theorem 2.5) gives $\mathbb{Z}/n \xrightarrow{m} \mathbb{Z}/n \rightarrow \mathbb{Z}/m \otimes \mathbb{Z}/n \rightarrow 0$, whose cokernel is $\mathbb{Z}/n / m(\mathbb{Z}/n) \cong \mathbb{Z}/(m, n)$. In particular $\mathbb{Z}/m \otimes \mathbb{Z}/m \cong \mathbb{Z}/m$ and $\mathbb{Z}/2 \otimes \mathbb{Z}/3 = 0$.

- $M \otimes_R R \cong M$ via $m \otimes r \mapsto rm$.
- $S^{-1}M \cong S^{-1}R \otimes_R M$ via $\frac{m}{s} \mapsto \frac{1}{s} \otimes m$.
- $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C}$: write $\mathbb{C} = \mathbb{R}[x]/(x^2 + 1)$; then $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[x]/(x^2 + 1) \cong \mathbb{C}[x]/(x - i) \times \mathbb{C}[x]/(x + i) \cong \mathbb{C} \oplus \mathbb{C}$. Geometrically (Figure 1): base-changing the “point” $\text{Spec } \mathbb{C} \rightarrow \text{Spec } \mathbb{R}$ splits it into *two* points — the two conjugate embeddings. Points multiply under base change!

Theorem 2.5 (Right-exactness). *If $A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ is exact and M is any module, then $A \otimes M \xrightarrow{f \otimes 1} B \otimes M \xrightarrow{g \otimes 1} C \otimes M \rightarrow 0$ is exact.*

Proof. $g \otimes 1$ is surjective because g hits a representative of every simple tensor. $(g \otimes 1)(f \otimes 1) = (gf) \otimes 1 = 0$ so $\text{im}(f \otimes 1) \subseteq \ker(g \otimes 1)$. For the reverse, let $Q = (B \otimes M) / \text{im}(f \otimes 1)$ and define $\beta : C \times M \rightarrow Q$ by $\beta(c, m) = \bar{b} \otimes m$ where $g(b) = c$; independence of the lift b holds because two lifts differ by $f(a)$, whose tensor lies in $\text{im}(f \otimes 1)$, i.e. is 0 in Q ; bilinearity is clear, so β induces $\bar{\beta} : C \otimes M \rightarrow Q$. The quotient map $B \otimes M \rightarrow C \otimes M$ factors through Q and the two maps are mutually inverse; hence $\ker(g \otimes 1) = \text{im}(f \otimes 1)$. $\square$

Example 2.6 (Tensor is *not* left-exact). Tensor $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow \mathbb{Z}/2 \rightarrow 0$ with $\mathbb{Z}/2$: since $\times 2$ becomes 0 on $\mathbb{Z}/2$, we get $\mathbb{Z}/2 \xrightarrow{0} \mathbb{Z}/2 \rightarrow \mathbb{Z}/2 \rightarrow 0$, which fails exactness at the left. The kernel that tensoring *lost* is precisely what Tor will recover in S43; modules for which no information is ever lost are the **flat** modules (§4).

3 Nakayama’s Lemma

Theorem 3.1 (Nakayama). *Let $(R, \mathfrak{m})$ be local and M a finitely generated *R*-module. If $\mathfrak{m}M = M$ then $M = 0$.*

fiber = $\text{Spec}(S \otimes_{\mathbb{R}} k(\mathfrak{p}))$: here two points

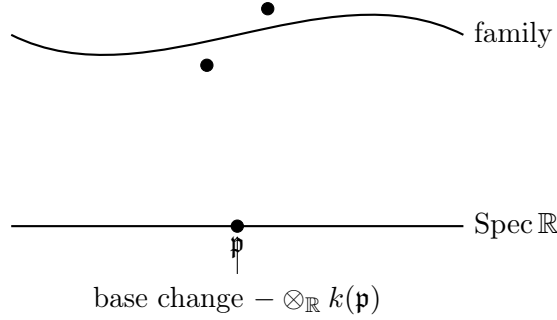


Figure 1: Tensoring with the residue field $k(\mathfrak{p})$ extracts the fiber of a family over the point $\mathfrak{p}$ — evaluation of a module at a point.

Proof (determinant trick). Let $M = Rm_1 + \cdots + Rm_n$. From $\mathfrak{m}M = M$ write $m_i = \sum_j a_{ij}m_j$ with $a_{ij} \in \mathfrak{m}$. With $A = (a_{ij})$ and I_n the identity matrix, $(I_n - A) \cdot (m_1, \dots, m_n)^T = 0$. Multiplying by the adjugate of $I_n - A$ gives $\det(I_n - A)m_i = 0$ for all i , hence $\det(I_n - A)M = 0$. Expanding the determinant, $\det(I_n - A) = 1 + u$ with $u \in \mathfrak{m}$; in a local ring $1 + u$ is a unit (else $1 \in \mathfrak{m}$). A unit annihilating M forces $M = 0$. $\square$

Corollary 3.2. *If $N \subseteq M$ with $M = N + \mathfrak{m}M$ then $N = M$. (Apply Theorem 3.1 to M/N .)*

Corollary 3.3 (Generators lift). *If $m_1, \dots, m_n \in M$ have images spanning the $R/\mathfrak{m}$ -vector space $M/\mathfrak{m}M$, then $m_1, \dots, m_n$ generate M . Consequently the minimal number of generators of M equals $\dim_{R/\mathfrak{m}}(M/\mathfrak{m}M)$.*

Proof. Take $N = Rm_1 + \cdots + Rm_n$; then $M = N + \mathfrak{m}M$, so Corollary 3.2 gives $M = N$. Minimality: any generating set maps to a spanning set of $M/\mathfrak{m}M$, and a basis of $M/\mathfrak{m}M$ lifts to a minimal generating set by the first sentence. $\square$

Remark 3.4 (Why Nakayama matters here and later). $M/\mathfrak{m}M$ is the *fiber* of M at the point $\mathfrak{m}$ (Figure 1); Nakayama says a f.g. module is controlled by its fiber. This is the engine behind: generators of stalks of sheaves (S26–S28), tangent spaces and smoothness (S30), and the equivalence “flat + f.g. over local = free” that makes flatness the algebraic form of “vector bundles vary continuously” (S30).

4 Flatness

Definition 4.1 (Flat module). M is **flat** if $M \otimes_R -$ preserves exactness of every s.e.s.; equivalently, for every injective $A \hookrightarrow B$, the map $A \otimes M \hookrightarrow B \otimes M$ is injective.

Example 4.2. • Free modules are flat (tensor with R^n is a direct sum of copies).

- Direct sums and direct limits of flat modules are flat.
- $S^{-1}R$ is flat over R : localization is exact (Proposition 4.3).
- $\mathbb{Z}/n$ is *not* flat over $\mathbb{Z}$ (Example 2.6 with $M = \mathbb{Z}/n$).
- Fields are flat; more generally any localization is (previous item).

Proposition 4.3 (Localization is exact, hence flat). *For a multiplicatively closed $S \subseteq R$, the functor $M \mapsto S^{-1}M$ is exact; in particular $S^{-1}R$ is a flat R -module, and $S^{-1}M \cong S^{-1}R \otimes_R M$.*

Proof. The isomorphism $\frac{1}{s} \otimes m \leftrightarrow \frac{m}{s}$ is well-defined by the universal property (Theorem 2.3). Exactness: surjectivity and $\text{im} \subseteq \ker$ are formal; for injectivity, if $\frac{a}{s} \in S^{-1}A$ maps to 0 in $S^{-1}B$ then $t f(a) = 0$ in B for some $t \in S$, hence $f(ta) = 0$ so $ta \in \ker f = 0$, whence $\frac{a}{s} = \frac{ta}{ts} = 0$ in $S^{-1}A$. $\square$

Remark 4.4 (Preview: flatness in S30 and in homological algebra). Over a local ring, *flat* + *finitely presented* = *free*; globally, *flat* + *f.g.* = *locally free* (a vector bundle). Étale maps (S30) are *flat* + *unramified*: geometrically “local isomorphisms”, algebraically “fibers are finite sets of reduced points”. Homologically, flatness is the vanishing of Tor (S43): M *flat* $\iff \text{Tor}_1^R(M, N) = 0$ for all N . Thus this session’s right-exactness is the seed of the entire derived-functor machine of S42–S45.

5 Exercises

Exercise 1. Prove $\mathbb{Z}/m \otimes_{\mathbb{Z}} \mathbb{Z}/n \cong \mathbb{Z}/(m, n)$ directly from the universal property (Theorem 2.3), without using right-exactness.

Exercise 2. Show $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q} = 0$. Conclude that $\mathbb{Q}/\mathbb{Z}$ is neither flat nor free over $\mathbb{Z}$.

Exercise 3. Prove Theorem 2.5 in full detail, including the well-definedness of β .

Exercise 4. Using Example 2.6, exhibit an injective map $A \hookrightarrow B$ such that $A \otimes \mathbb{Z}/2 \rightarrow B \otimes \mathbb{Z}/2$ is *not* injective.

Exercise 5. Prove the splitting lemma (Proposition 1.9) in all directions, and give a s.e.s. of $\mathbb{Z}$ -modules that does *not* split.

Exercise 6. Reprove Nakayama (Theorem 3.1) for $n = 1$ and $n = 2$ generators by hand, then deduce Corollaries 3.2 and 3.3 from the general case.

Exercise 7. Let $(R, \mathfrak{m})$ be local and M f.g. with $M/\mathfrak{m}M = 0$. Show $M = 0$. Give an example of a *non*-f.g. module over a local ring with $M = \mathfrak{m}M \neq 0$ (hint: $M = R/\mathfrak{m}$ -vector spaces of infinite dimension won’t work; try $M = \mathbb{Q}$ over $\mathbb{Z}_{(p)}$... check carefully).

Exercise 8. Prove $M \otimes_R R/I \cong M/IM$ for any ideal I , and interpret it as “restriction of the family to the closed subset $V(I)$ ”.

Exercise 9. Show $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C}$ explicitly (idempotents $e_{\pm} = \frac{1}{2}(1 \otimes 1 \mp i \otimes i)$), and identify the two points of $\text{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C})$.

Exercise 10. Prove that $S^{-1}R$ flat over R implies: if $M \hookrightarrow N$ is injective then $S^{-1}M \hookrightarrow S^{-1}N$ is injective. Conversely, show $\mathbb{Z}_{(p)}$ is flat over $\mathbb{Z}$ but $\mathbb{Z}/p$ is not.

Exercise 11. Let M, N be f.g. modules over a local ring $(R, \mathfrak{m})$. If $M \otimes_R N = 0$, prove $M = 0$ or $N = 0$ (hint: look at fibers $M/\mathfrak{m}M$, $N/\mathfrak{m}N$ and use Nakayama).

Exercise 12. Show that the rank of a free module over a commutative ring is well defined (Proposition in §1), and deduce that $R^m \cong R^n \Rightarrow m = n$ even when R has zero-divisors.

Where This Session Feeds the Course

- **Depends on:** S1 (rings, ideals, localization, Krull for maximal ideals).
- **Feeds:** S24 (fractional ideals as modules); S26–S28 (sheaves of modules, stalks, fibers $M \otimes k(\mathfrak{p})$); S30 (flat/étale/smooth, Jacobian criterion); S32 (Kähler differentials); S42–S45 (Tor, Ext, cohomology: the derived repair of right-exactness).
- **Unifying thread:** a module is a family of vector spaces over the points of $\text{Spec } R$; tensoring with $k(\mathfrak{p})$ evaluates the family at $\mathfrak{p}$; base change $S \otimes_R -$ pulls the family back along $\text{Spec } S \rightarrow \text{Spec } R$; flatness says the fibers vary without jumps.

References for this session

- [1] Atiyah, M.F. & Macdonald, I.G. *Introduction to Commutative Algebra*, ch.2–3.
- [2] Eisenbud, D. *Commutative Algebra with a View Toward Algebraic Geometry*, ch.1, 6–7.
- [3] Lang, S. *Algebra*, ch.XVI.
- [4] Matsumura, H. *Commutative Ring Theory*, ch.2–3 (Nakayama, flatness).

End of Session S2 — next: Session S3, Fields, Extensions, and Algebraic Closure